Symplectic manifold

- M manifold
- closed non-degenerate 2-form ω
- we use ω to define vector fields from functions

$$f \in \mathcal{C}^\infty(M, \mathbb{R}) \mapsto X_f \in \mathfrak{X}(M)$$

A symplectic form on M is closed, non-degenerate 2-form ω

$d\omega = 0$

$\omega(u, v) = 0 \quad \forall v$

$\Rightarrow u = 0$

- note: since ω antisymmetric, non-degenerate implies $\dim(M)$ is even

associate locally a matrix with ω

$\hookrightarrow \quad \omega|_a : T_aM \times T_aM \to \mathbb{R}$ bilinear antisymmetric

choose a basis of T_aM $\{e_\mu\}$

$\Omega_{\mu\nu} = \omega|_a(e_\mu, e_\nu)$ consider $\Omega_{\mu\nu}$ as elements of a matrix $(\dim(M) \times \dim(M))$

$\Omega_{\nu\mu} = -\Omega_{\mu\nu}$

$\omega|_a$ non-degenerate $\Leftrightarrow$ $\det \Omega \neq 0$

$$\det(\Omega) = \det(\Omega^t) = \det(-\Omega) = (-1)^{\dim(M)} \det(\Omega)$$

Theorem (Darboux) if (M, ω) is a symplectic $2n$-manifold then for each point $a \in M$ there is a local chart (U, φ)

$\varphi = (q^1, q^2, \dots, q^n, p_1, \dots, p_n)$

s.t $\boxed{\omega|_U = dq^\mu \wedge dp_\mu}$

$\rightarrow$ locally all symplectic manifolds (of same dimension) are diffeomorphic

S set and $|S| = |\mathbb{R}|$ $\mathbb{R}^2$ can be a 1-dimensional manifold

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$\boxed{\text{examples}}$ $M = \mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ global coordinates $(q^1, q^2, q^3, \dots, q^n, p_1, p_2, \dots p_n)$

$$\omega = dq^\mu \wedge dp_\mu \qquad d\omega = \underbrace{d^2 q^\mu}_{=0} \wedge dp_\mu - dq^\mu \wedge \underbrace{d^2 p_\mu}_{=0} = 0$$

what is associated matrix ?

basis of tangent space $\left\{ \frac{\partial}{\partial q^1}, \frac{\partial}{\partial q^2} \dots \frac{\partial}{\partial p_1} \dots \frac{\partial}{\partial p_n} \right\}$

$$\Omega = \left(\begin{array}{c|c} 0 & \mathbb{1}_n \\ \hline -\mathbb{1}_n & 0 \end{array} \right) \qquad \det(\Omega) = 1$$

<u>ω exact</u> : $\omega = -d(\underbrace{p_\mu dq^\mu}_{\theta \text{ canonical 1-form}}) = -dp_\mu \wedge dq^\mu = dq^\mu \wedge dp_\mu$

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ex 2 Q any manifold look at $M = T^*Q$ $(\dim M = 2 \dim(Q))$

T^*Q has a canonical 1-form

$q^\mu : U \to \mathbb{R}$

give coordinates to T^*Q locally : on $U \subseteq Q$ coordinates $(q^1, \dots, q^n)$

extend them to T^*Q with coordinates $(q^1, \dots, q^n, p_1, \dots p_n)$

$\to$ coordinates on T^*Q

for a point $(a, \alpha) \in T^*Q$ $(\alpha \in T^*_a Q)$

we define $q^\mu(a, \alpha) = q^\mu(a)$

coordinate basis of $T^*_a Q$

for p_μ we the fact that $\alpha = \alpha_\mu dq^\mu_a$

$\to p_\mu(a, \alpha_\nu dq^\nu_a) = \alpha_\mu$

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define locally $\theta = p_\mu dq^\mu$ $\to$ section of $T^*(T^*Q)$

$\to$ 1-form on T^*Q

θ does not depend on choice of chart on Q

if we change coordinates on Q (use $\tilde{q}^\mu$)

$$\tilde{P}_\mu(a, \tilde{\alpha}_\nu d\tilde{q}^\nu_a) = \tilde{\alpha}_\mu$$

$$\underbrace{\alpha}_{\in T^*_a Q} = \overset{\in \mathbb{R}}{\alpha_\mu} \underbrace{dq^\mu_a}_{\in T^*_a Q} = \overset{\in \mathbb{R}}{\tilde{\alpha}_\nu} \underbrace{d\tilde{q}^\nu_a}_{\in T^*_a Q}$$

what is $\tilde{\alpha}_\mu$ in terms of α_μ

$$\underbrace{df_a}_{\text{pushforward}}(v) = \overset{\text{derivation}}{v} f$$

$$\alpha\left(\frac{\partial}{\partial q^\mu}\Big|_a\right) = \alpha_\mu = \tilde{\alpha}_\nu \, d\tilde{q}^\nu_a\left(\frac{\partial}{\partial q^\mu}\Big|_a\right)$$

$$= \tilde{\alpha}_\nu \frac{\partial}{\partial q^\mu}\Big|_a \tilde{q}^\nu$$

$$\boxed{d\tilde{q}^\nu = \frac{\partial \tilde{q}^\nu}{\partial q^\mu} dq^\mu}$$

$$= \tilde{\alpha}_\nu \frac{\partial \tilde{q}^\nu}{\partial q^\mu}(a)$$

Section of change of coordinates

$\varphi_1 : (0,1) \to \mathbb{R}$, $x \mapsto x$

$\varphi_2 : (0,1) \to \mathbb{R}$, $x \mapsto \ln(x)$

$\mathrm{Im}\,\varphi_2 = (-\infty, 0)$

$$P_\mu(a, \alpha) = P_\mu(a, \tilde{\alpha}_\nu d\tilde{q}^\nu_a) = P_\mu\left(a, \tilde{\alpha}_\nu \frac{\partial \tilde{q}^\nu}{\partial q^\sigma} dq^\sigma_a\right)$$

$$= \tilde{\alpha}_\nu \frac{\partial \tilde{q}^\nu}{\partial q^\mu} = \frac{\partial \tilde{q}^\nu}{\partial q^\mu} \tilde{P}_\nu(a, \alpha)$$

$$P_\mu = \frac{\partial \tilde{q}^\nu}{\partial q^\mu} \tilde{P}_\nu$$

$$P_\mu dq^\mu = \frac{\partial \tilde{q}^\nu}{\partial q^\mu} \tilde{P}_\nu dq^\mu = \tilde{P}_\nu d\tilde{q}^\nu$$

Coordinate free

$\pi : T^*Q \to Q$, $(a, \alpha) \mapsto a$

$d\pi_a : \overbrace{T_{(a,\alpha)}(T^*Q)}^{\dim = 4 \times \dim(Q)} \to T_a Q$

$(a, \alpha) \in T^*Q$

$\alpha : T_a Q \to \mathbb{R}$

$\theta_{(a,\alpha)} := \alpha \circ d\pi_{(a,\alpha)} : T_{(a,\alpha)}(T^*Q) \to \mathbb{R}$ linear

$\theta_{(a,\alpha)} \in T^*_{(a,\alpha)}(T^*Q)$

$M = T^*Q$

$\theta : M \to T^*M$, $m \mapsto (m, \theta_m)$

$$\begin{tikzcd} T_{(a,\alpha)}(T^*Q) \arrow[dr, "d\pi_{(a,\alpha)}"] \arrow[dr, phantom, shift right=4ex, "\theta_{(a,\alpha)}"] & \\ \mathbb{R} & T_a Q \arrow[l, "\alpha"'] \end{tikzcd}$$

if we try some for $M = TQ$ $\qquad \pi : TQ \to Q$

$d\pi_m : T_m(TQ) \to T_{\pi(m)} Q$

if $m = (q, v) \in TQ$

v is a vector

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in coords $\qquad \theta = p_\mu \, dq^\mu = -q^\mu \, dp_\mu$ + closed form

define ω from θ : $\boxed{\omega = -d\theta}$ canonical symplectic form

- $d\omega = 0$ since ω exact
- ω non-degenerate because locally $\omega = -d(p_\mu \, dq^\mu) = dq^\mu \wedge dp_\mu$ which is non-degenerate

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Hamiltonian vector fields

if $f \in \mathcal{C}^\infty(M, \mathbb{R})$ $\qquad \omega$ symplectic

Hamiltonian vector field associated to f (X_f) define by

$$\boxed{\omega(X_f, \cdot) = df}$$

$\leftarrow$ to find X_f (f and ω are given)

aside: if g is a pseudo-Riemannian metric tensor and $f \in \mathcal{C}^\infty(M, \mathbb{R})$ define $\operatorname{grad} f$ by

$$g(\operatorname{grad} f, \cdot) = df$$

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in Darboux coordinates

$\omega = dq^\mu \wedge dp_\mu \qquad X_f = A^\mu \frac{\partial}{\partial q^\mu} + B_\mu \frac{\partial}{\partial p_\mu}$

$$\omega(X_f, \cdot) = A^\mu \, dp_\mu - B_\mu \, dq^\mu$$

$$\| $$

$$df = \frac{\partial f}{\partial q^\mu} dq^\mu + \frac{\partial f}{\partial p_\mu} dp_\mu$$

$\to \quad A^\mu = \frac{\partial f}{\partial p_\mu} \qquad B_\mu = \frac{\partial f}{\partial q^\mu}$

$\iota_{X_f} \omega$

$$\boxed{X_f = \frac{\partial f}{\partial p_\mu} \frac{\partial}{\partial q^\mu} - \frac{\partial f}{\partial q^\mu} \frac{\partial}{\partial p_\mu}}$$

Dynamics

$\gamma : I \subseteq \mathbb{R} \to M$ curve find diff. equation

choose a special function H (Hamiltonian) ← smooth

$$\begin{cases} \dot\gamma(t) = X_H|_{\gamma(t)} \\ \gamma(0) = a \end{cases} \to \text{solution exists and is unique}$$

→ Hamilton's equation

in local Darboux coordinates $\omega = dq^\mu \wedge dp_\mu$

$$X_H = \frac{\partial H}{\partial p_\mu}\frac{\partial}{\partial q^\mu} - \frac{\partial H}{\partial q^\mu}\frac{\partial}{\partial p_\mu} \qquad \gamma(t) = (q^1(t), q^2(t), \dots, p_1(t), \dots, p_n(t))$$

$$\dot\gamma(t) = \dot q^\mu(t) \frac{\partial}{\partial q^\mu}\Big|_{\gamma(t)} + \dot p_\mu(t) \frac{\partial}{\partial p_\mu}\Big|_{\gamma(t)} \qquad \Big| \qquad \dot\gamma(t) f = \frac{d}{ds} f(\gamma(s))\Big|_{s=t}$$

$$\to \begin{cases} \dot q^\mu = \dfrac{\partial H}{\partial p_\mu} \\ \dot p_\mu = -\dfrac{\partial H}{\partial q^\mu} \end{cases}$$

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to get Schrodinger eq:

$M = \mathbb{C}^n$ coords $z, \bar z$

$$h = d\bar z_\mu \otimes dz^\mu$$

$$\omega = -i\hbar(h - \bar h) = -i\hbar\, d\bar z_\mu \wedge dz^\mu$$

if $f : \mathbb{C}^n \to \mathbb{R}$

X_f defined by $\omega(X_f, \cdot) = df$

$$X_f = A^\mu \frac{\partial}{\partial z^\mu} + B_\mu \frac{\partial}{\partial \bar z_\mu} \qquad \omega(X_f, \cdot) = -i\hbar\left(B_\mu dz^\mu - A^\mu d\bar z_\mu\right)$$

$$\| $$

$$df = \frac{\partial f}{\partial z^\mu} dz^\mu + \frac{\partial f}{\partial \bar z_\mu} d\bar z_\mu$$

$$X_f = \frac{1}{i\hbar}\left(\frac{\partial f}{\partial \bar{z}_\mu}\frac{\partial}{\partial z^\mu} - \frac{\partial f}{\partial z^\mu}\frac{\partial}{\partial \bar{z}_\mu}\right)$$

$\mathcal{H}: \mathbb{C}^n \to \mathbb{R}$ Hamiltonian function

Hamiltonian operator

$$\mathcal{H}(z,\bar{z}) = z^* \hat{H} z \longrightarrow \text{eqiv. to } \langle\psi|H|\psi\rangle$$

$$= \bar{z}_\mu H^\mu{}_\nu z^\nu$$

$$X_{\mathcal{H}} = \frac{1}{i\hbar} H^\mu{}_\nu z^\nu \frac{\partial}{\partial z^\mu} - \frac{1}{i\hbar} \bar{z}_\nu H^\nu{}_\mu \frac{\partial}{\partial \bar{z}_\mu}$$

$$\dot{\gamma} = X_{\mathcal{H}} \qquad \dot{\gamma} = \dot{z}^\mu \frac{\partial}{\partial z^\mu} + \dot{\bar{z}}_\mu \frac{\partial}{\partial \bar{z}_\mu}$$

$$\to \dot{z}^\mu = \frac{1}{i\hbar} H^\mu{}_\nu z^\nu \qquad |\psi\rangle = \begin{pmatrix} z^1 \\ \vdots \\ z^n \end{pmatrix}$$

c.c.

$$\to i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$$